

Computer Systems Security

2025/26

Bachelor in Informatics Engineering
University of Minho

Vítor Francisco Fonte, vff@di.uminho.pt, University of Minho, 2025-26

Teaching Team

Name	E-mail	Classes	Office	Off-class sessions (must be scheduled in advance)
Vítor Francisco Fonte	vff@di.uminho.pt	T1	2.16	5a-f, 12-13h
Hugo José Pacheco	hpacheco@di.uminho.pt	T1, PL4, PL5	2.15	2a-f, 11-12h
João Marco Cardoso Silva	joaomarco@di.uminho.pt	PL1	2.16	4a-f, 14-15h
Diogo Pinto Ribeiro	d13847@di.uminho.pt	PL2, PL3	2.12	4a-f, 11-12h

Theoretical Classes

Syllabus

- Security fundamentals
- Identity, authentication and access control
- Symmetric and asymmetric cryptography
- Cryptographic signatures, PKI, and applications
- Software security
- Web security
- Operating system security
- Network security

Laboratory Classes

- Laboratory guides/scripts:
 - Following the contents covered in the theoretical component
- You will need a computer with a basic Linux operating system:
 - Usage of a virtual machine (VM) is highly recommended
 - Command-line interface (CLI) installation (no GUI) is sufficient
 - Note: A lab setup guide will be released shortly

Assessment

Final mark

- **Individual written test: May 21, 55%**
 - Minimum of 8 points out of 20 is required to pass
- **Group project assignment: May 23, 35%**
 - A minimum of 10 points is required to pass and qualify for a resit exam
 - Group assignment, individual assessment based on each student's contribution
- **Continuous assessment: 10%**
 - Based on individual contribution to class activities and lab assignments
- **Important note:**
 - If deemed necessary by the teaching staff, the mark awarded for the written test or the practical project may have to be defended in a subsequent oral examination

Assessment

Project assignment

- Groups of three students (more information to follow)
- Submission via GitHub (private group repository)
- Deadline is May 23, discussion in June 1 to 5
- Late submissions incur a 10% penalty per day (granularity is 24 hours)

Assessment

Straightforward practicals in lab classes

- Individual submission via GitHub (private individual repository)
- No formal grading but may be used later on for continuous assessment
- Submission date to be announced with each lab guide

Engaging with Lecturers

- **Do you have questions regarding a topic?**
 - Please ask them during the relevant lecture/lab or at the very next class
- **All other communication should be carried out via Blackboard**
 - E.g. scheduling a meeting with a lecturer, clarification on course logistics, etc
 - A response will typically be provided within three working days

Bibliography

- Crypto 101, <https://www.crypto101.io/>
- “Computer Security, Principles and Practice”, William Stallings, 4th Ed., 2018.
- “Computer Security”, Dieter Gollmann, 3rd Ed, 2011.
- “Computer Security: A Hands-On Approach”, Wenliang Du, 3rd Ed., 2022.

Course Activation in Blackboard

- Students without current access to the module on Blackboard should use the activation code below to enrol.
 - LEISSI2025
- Please share it with any fellow students who require access to this module.